

Gavin Precurato

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EDUCATION

Youngstown State University – Rayen School of Engineering (ABET Accredited)

Expected May 2028

Bachelor of Engineering in Mechanical Engineering

Relevant Courses: Strength of Materials, Engineering Communication with CAD, Differential Equations, Dynamics, Physics 2.

SKILLS

Engineering: Design for Additive Manufacturing, (DFAM) Design for Manufacturing, (DFM) Rapid Prototyping, Technical Drawings, GD&T, Composite Layups, Control Systems/PID.

Software/Programming: Siemens NX, SolidWorks, AutoCAD, OnShape, Microsoft Office, Java, Python, C, MATLAB, Cura.

EXPERIENCE

Mechanical Engineering Intern, Center Street Technologies

May 2025 – May 2026

Youngstown, OH

- Engineered and deployed large scale advanced manufacturing solutions utilizing 3D printing, scanning, machining, and rheology to produce precision tooling at an industrial scale.
- Designed and fabricated over 15 fixtures and structures to improve the safety, setup efficiency, and consistency of manufacturing processes.
- Supported implementation of ISO 9001, AS9100D, and CMMC compliance procedures, gaining experience with relevant quality, reliability, and security guidelines used in aerospace and defense.
- Wrote and presented monthly reports to leadership, documenting enhancements to production capabilities.
- Interpreted electrical schematics to complete panel wiring, troubleshooting, and maintenance of equipment.

Design Engineering Intern, Vista AST

Jun 2024 – May 2025

Youngstown, OH

- Designed 4 interactive "Career Carts" for a variety of STEM jobs and trades, reaching over 3,000 students and exposing them to future career opportunities.
- Led groups of 10-35 students on field trips and summer camps with hands-on engineering activities and guided them through challenges while strengthening their problem solving abilities.

Team Captain & Mentor, F.I.R.S.T Robotics Team 3193

Dec 2019 – Current

- Designed and iterated subsystems for a 110 pound robot to accomplish game tasks within a 2-3 month timeline.
- Led the programming subteam, communicating effectively with design and assembly teams to establish full system robot integration and automation techniques.
- Diagnosed and resolved electromechanical failures under strict time constraints during competition events.
- Mentored teams of 15-20 students in the production and programming of high precision mechanisms using kinematics, vision processing, and closed loop feedback systems.

PROJECTS

NHRL Combat Robot, YSU Robotics Club

Dec 2025 – Apr 2026

- Designed a combat robot within a 30 pound weight constraint and selected materials based on their durability.
- Applied dynamics coursework to calculate energy transfer and speed of the robot's systems during competition.
- Designed components for both additive and traditional manufacturing techniques, enabling rapid prototyping and weight reduction of non-critical parts by up to 10% of the robot's overall weight.

3D Printed NERF Blaster

Jan 2025

- Designed an elastic band powered NERF blaster with over 93% additively manufactured components.
- Modeled full assembly in CAD to analyze movement, failure points, and tolerances.
- Executed rapid prototyping with 3D printing, reducing part testing time and improving validation speed from days to hours.